Conveyor and Test Bench Drives — Maintenance Guide

Belt conveyors and torque test benches

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Plant engineering

Read this manual before installing, operating or servicing the equipment. Keep it at the machine.

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1 Scope

This guide covers the belt conveyors and the end-of-line torque test benches. Both use small asynchronous motors (1.5–3 kW, 1,450 rpm) with shaft couplings.

2 Normal values

Machine	Vibration RMS	Bearing temperature	Power
Belt conveyor	1.3–1.9 mm/s	40–46 °C	2.5–3.5 kW
Torque test bench	0.9–1.3 mm/s	38–44 °C	2–3 kW

Small machines stay far below the 4.5 mm/s plant alarm even when they are badly damaged. Compare each machine with its own normal values instead (PS-VIB-01, section 3).

3 Conveyors

3.1 Belt tracking

A belt that runs off-centre loads one drum bearing and rubs the frame. Adjust the tail drum, 1/4 turn at a time. A steady rise of vibration with normal kurtosis points to tracking or drum misalignment.

3.2 Drum bearings

Drive drum bearings (SKF-6204) wear from belt tension and contamination. Rising kurtosis and bearing temperature mean damage; replace both bearings.

3.3 Motor overload

Jammed return rollers or a belt that is too tight overload the motor: current and temperature rise while vibration stays normal. Free the rollers, check the tension.

4 Test benches

4.1 Coupling

The coupling (CPL-19) between motor and torque sensor must be aligned within 0.05 mm. Misalignment falsifies torque readings and raises vibration.

4.2 Cooling

The bench motor has a separate fan (FAN-TT2). A failed fan makes the motor overheat in a few hours.

5 Maintenance plan

Interval	Conveyor	Test bench
Weekly	Tracking, belt surface	Fan, cables
Monthly	Drum bearings by ear and trend	Coupling check
6 months	Bearings, belt tension, motor	Coupling alignment, calibration